3.11 Curves

class 3 6 August 24

Curve
1. Horizontal curve
2. Vertical curve

- Horiz curve
- kahan deteh?

Horizontal curve (superelevation) : is provided in the section of road or pavement where change in direction is required.

- Horizontal curve ka introduction kya karta h.
  - radius ko
  - centrifugal force ko

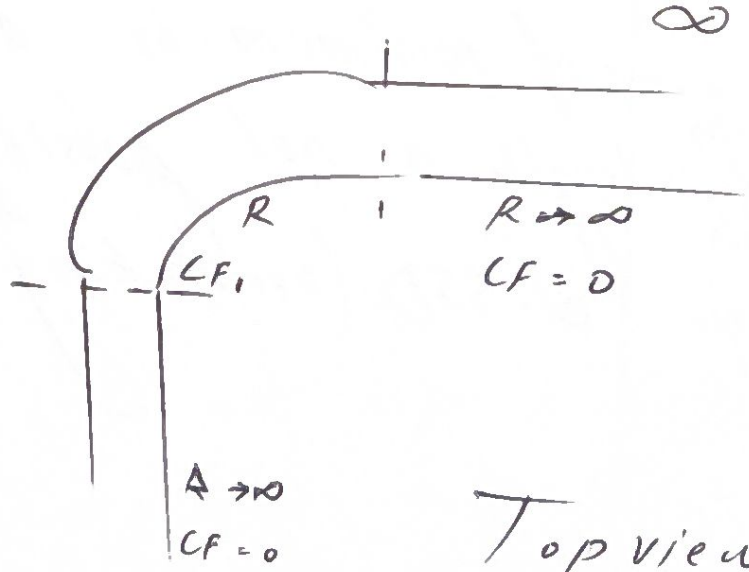
Introduction of horizontal curve gives -

① decreases radius from ∞ to R & then increases to ∞.

② gradual introduction of centrifugal force from 0 to some value then zero

0 - CF - 0
∞ - R - ∞

- Diagram.



Top view.

Horizontal curve & centrifugal force

- kis direction meh?
- kya cause karta h?
- formula

Definitions:

1) Centrifugal force on vehicle:

When a vehicle travels in a horizontal curve then an outward force acts on it which creates condition of vehicle to overturn or lateral skid

$$CF = P = \frac{Wv^2}{gR} = \frac{Wt \times vel^2}{gacc \times Rad}$$

W = wt of vehicle in kg

v is velocity of vehicle in ms^{-1}

R is radius of circular curve in metre.

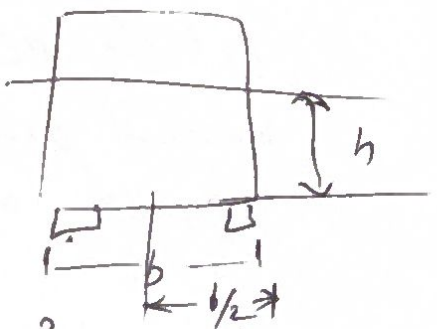
g is gravitational acceleration in ms^{-2}.

- Impact factor
  - alternate name
  - formula.

(alternate name)

2) Impact factor or centrifugal ratio!

$$\frac{P}{W} = \frac{v^2}{gR}$$



- Conditions
  - for ⇌
    - formula
    - b kyah
    - h kyah

Condition

1) for ⇌ $\frac{P}{W} = \frac{v^2}{gR} = \frac{b}{2h}$

b is width of vehicle &

h is height of CG of vehicle or ht of centrifugal force acting from road surface.

3.3 Conditions in horiz curve

(2) to avoid over turning

(3) to avoid lateral skid

(ii) to avoid overturning

$$\frac{b}{2h} > \frac{f}{W}$$

(iii) to avoid lateral skid

$$f_{lateral} > P/W$$

for the condition of sliding (skidding) and no overturning

$$f_{lateral} > P/W \quad \& \quad b/2h > P/W$$

- for no sliding & no overturning (formula)

- flat pavement me curve pe vehicle pe kahan pressure zyada ayega

- inner wheels pe pressure zero kab hoga.

Note: ① if the pavement is kept flat then the pressure on outer wheels of vehicle will be more than pressure on inner wheels due to centrifugal force.

② for limiting condition if overturning occurs then pressure @ inner wheel becomes equal to zero. (generally for flat roads)

Horiz curve is superelevn

— kya h?

Superelevation - e or cant

- Rise of outer edge of pavement wrt inner edge on horizontal curve it is called superelevation or cant or banking.

— Y verse h ki longitudnal?

- It is a transverse slope.

— e, θ se kaise related h?

- $e = \tan\theta = \frac{x}{y}$

① General equation with fric & superelvn

1) General equation (with friction & superelevation)

$$(e + f_{lateral}) = \frac{v^2}{gR}$$

② equilibrium superelevn formula

2) Equilibrium superelevation ($f_{lateral} = 0$)

$$e = \frac{v^2}{gR} = e_{eq} = \frac{V^2}{127R}$$

— flat kitna h?

for this condition coefficient of Lateral friction is considered zero.

③ allowable speed without superelevn v formula

3) allowable speed of vehicle on a horiz curve without superelevation. ($e = 0$)

$$f_{lateral} = \frac{v^2}{gR} \Rightarrow \boxed{V = \sqrt{f_{lat} gR}}$$

$V_{allowable}$ = speed

Superelevn design steps

(ii) If e > emax toh kya karenge

then superelevation should be provided e. (chote wale ko denge) (3.6)

emax ki value IRC se milta hai.

(iii) if e > emax (chote wale ko denge)

then provide superelevation emax.

Step ③ →

Step ③

- f calc.

Calculate friction from general eqn

$$e + f_{lat} = \frac{V^2}{127R} = \frac{v^2}{gR}$$

$$f_{lat} = \frac{V^2}{127R} - e$$

- flat < 0.15

(i) if $f_{lat} < 0.15$ (codal value) OK (Chote wale ko denge)

put road friction as flat calculated in step 3

- flat > 0.15

(ii) if $f_{lat} > 0.15$ Not OK (Chote wale ko denge)

if not ok the speed should be restricted & provide flat = 0.15

- Step ④

3.5 Horiz curve

Superelevation

- Mixed traffic or design super elevation.
  - Reduced speed kitni lete h.
  - e formula
    - in ms^{-1}
    - in kmph

4) Superelevation for mixed traffic condition or design superelevation

$1v \longrightarrow 0.75v$

$e + f_{lat} = \frac{v^2 \longrightarrow 0.75v}{gR}$

$e = \frac{(0.75v)^2}{gR}$

$e = \frac{V^2 \longrightarrow kmph}{225R}$

$f_{lateral}$ ki value generally kya lete h!

$f_{lateral} = 0.15$ lete h. in general case @ me.

In case of mixed traffic condition the speed of vehicle remains 75% of its original speed. ie $\frac{3}{4}v$ or $.75v$

Steps

Design of superelevation

- Step ① - mixed traffic
  - e
    - general traffic
- Step ②
  - Condns
    - $e \leq e_{max}$

Design of superelevation steps (mixed traffic)

Step ①: calculate $e = \frac{v^2}{225R}$

general $e = \frac{v^2}{127R}$

Step ②:-

(i) If $e < e_{max}$

3.1) Design of superelevation

Step 4 v calculate

- restricted speed kitni denge.

Step 4

$e_{.....} + f_{int} = \frac{V^2}{127R}$

$e_{step\,②} + 0.15 = \frac{V^2}{127R}$

$\sqrt{127R(e+0.15)} = V_{restricted\ speed}$

Determine of radius

- kyah, kaise derive karte hai
- kyaus eh
- formula

Determinn of radius

① Ruling minimum Radius

Gl can be derived by maximum superelevation & maximum coeff of lateral friction. With design speed.

Ideally the radius of curve should be more than ruling radius

$e_{max} + f_{lat\,max} = \frac{V^2}{127 R_{Ruling}}$

IRC 3.15 IRC

Determin of radius

② abs min radius

- kaise derive karteh.

- formula.

V is design speed of vehicle.

① absolute minimum radius. (R_{abs})

Absolute minimum radius is obtained @ maximum superelevation & maximum coefficient of friction with minimum speed.

f_{max} + f_{latera} : $\frac{V_{min}^2}{127 R_{abs}}$

IRC 0.15

IRC

Horizontal curve extra widening design

- extra widening kyaheteh

- psychological

- kyaheteh

Horizontal Curve [extra widening]

. Extra widening on horizontal curve refers to additional width of carriageway ie required on curve from psychological aspect & mechanical aspect.

Psychological widening

. There is a tendency for the drivers to drive close to the outer edges of pavement

3.9 Extra widening

- Psychological widening
- formula.

hence some clearance or space is to be provided for crossing and overturning opens

$$Wps = \frac{V}{9.5\sqrt{R}}$$

R is radius of circular curve in m

V is velocity in kmph

W is obtaind in m.

- Off tracking
- formula
- kyo hotiki
- kiske liye defined ki?

Off tracking

formula

off tracking $= \left(\frac{l^2}{2R}\right)$

when a vehicle negotiates horizontal curve the rear wheel follows a path of shorter radius than front wheel, this phenomena is called off tracking.

- off tracking is defined for single lane

Extra widening
- Mechanical widening
  - formula
    - nkyas
    - lkyas
- Combined formula
- Sharp curve pe kahan dete h?
- Normal curve pe kaha dete h?

3.10

(3) Mechanical widening

W_m : no of lanes × ff tracking

$$: \frac{n l^2}{2R}$$

$$W_m = \frac{n l^2}{2R} \times m$$

(nos → n, m)

l is length of rigid wheel base

n is no of lanes of largest vehicle

$l = 6.1\,m$

(4) Total extra widening

$$W_{extra} = W_m + W_{ps}$$

$$W_{extra} = \frac{n l^2}{2R} + \frac{V}{9.5\sqrt{R}} \quad (V \rightarrow kmph)$$

W_{extra} ---- m

Note : on sharp curves extra widening is provided fully on inner side.

- on normal road extra widening is given equally on both sides at uniform rate.

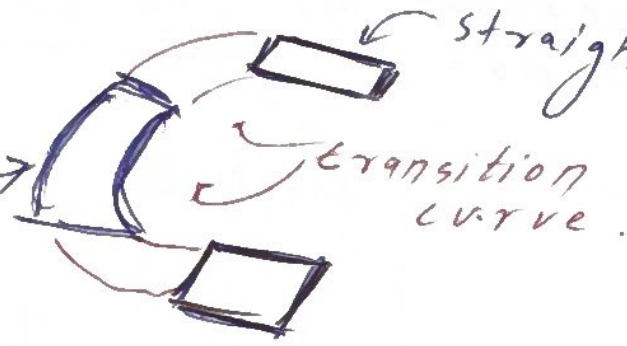
Horizontal curve design

- alternate name
  - transition curve milb kya hai
  - Transn curve kahan dete hai
- Diagram

Horizontal curve or Transition curve design.

Transition curve ka matlab highway engg me horizontal curve ko bola jata h. ↕ a little contradictory

Transition curve is a curve which is introduced between circular curve and straight line.



- Types
  - Spiral
    - alternate names
    - Speciality
    - length Radius relation proportionality
    - rate of chg of acc kaisa hai

Types / Shaps

1. Spiral or clothoid or glovers Spiral

- Ideal shape for transn curve
- LR $L \propto R^{-1}$
- rate of change of acceleration is uniform.

- Curve
  - types
    - Bernoulli's lemniscate
      - Used where?
      - Radius length relation.
      - Speciality
    - Cubic parabola
      - alternate name
      - used where
      - formula
    - Cubic Spiral
      - speciality

(ii) Bernoulli's lemniscate

- Mostly used in modern roads where deflection angle in curve is large
- Radius of curve decreases more rapidly with length.
- It is an autogenous curve.

(iii) Cubic parabola or Froude's easement curve

- used for valley curve
- $y = L^3 / 6RL_s$ L_s is length of SSD

(iv) Cubic spiral

- simple to set curve

4.1) kyo transition curve defe h²

- Grad in the
- steerg
- c

ngth of transn curve

- Rate of Δ of centr acc
  - formula
    - Cinmi
      - rang e of c

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Objecl of pooriding transilion curve

. to introduce gradually centrifuga l force between the tangent points and beginning of circular curve to avoid jerk.

. to enalle the driver to turn the steering gradually for comfort and security

. it introduces the superelevation and extra widening on curve gradually

Limiting length of transition curve.

... of.

rate of change of radial or centrifugal acceleration.

$L_s = \frac{V^3}{CR}$

V → velocity in ms⁻¹

R → radius of curve m

C → rate of change of centrifugal acceleration

length of transn curve

- c formula

$$C = \frac{80}{75 + V}$$

C → in m/s^3 ; V → in kmph

- c range

$-0.5 \leq c < 0.5$ ← range of c.

② Rate of Δ^{ge}

Rate of change of superelevation.

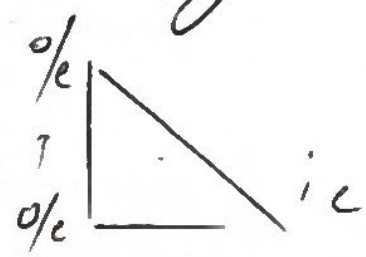
(i) rot abt inner edg

(i) if pavement is rotated about inner edge (rotated outer edge)

- formula

$$L_s = e N \left[W + W_e \right]$$

e → superelvn ; N → rate ; W → width ; W_e → extra widening



(ii) rot about centre line

(ii) When pavement is rotated about centre line (rotated inner & outer edge)

formula $L_s = e N \dfrac{\left[W + W_e \right]}{2}$



e is superelevation

length = superelevn × rated chg of selevn × [pvmt width + extra widening / 2]

We is extra widening

1 : N or 1 in N : rate of change of superelevation

A road 10m wide is deflected through an angle of 65° with centre line radius 350m a transition curve is used at each end of circular curve such that the gain in rated change of radial acceleration is 0.4 ms^{-3}. When the speed is 60 kmph find length of transition curve and shift

Ans given data

$V = 60\ kmph = \frac{60 \times 1000\ m}{60 \times 60\ sec} = 16.67\ ms^{-1}$

$\theta = 65°$

$C = 0.4\ m/sec^3$

$R = 350\ m$

$L_s = \frac{Velocity^3}{Change\ in\ rad} \times Radius$

$= \frac{(16.6)^3}{0.4 \times 350} = 33.07\ m.$

$S = \frac{Length\ of\ Xn\ curve^2}{24 \times Radius} = \frac{33.07}{24 \times 350}$

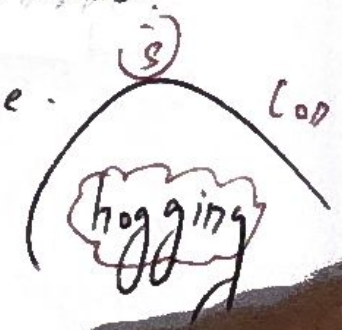
- Vertical curve
  - Provided where?
  - Summit curve
    - alternate names A, h, C

Vertical Curve

- Provided in the elevation @ change of gradient (level or vertical level)

1) Summit curve

- is called (check) ascending curve
- also called hogging curve.
- Convex curve



4.3 Length of Xs curve

- N value in plain & rolling terrain
- N value in hilly & mountainous

N as per IRC

N: 150 in plain and rolling terrain

N: 60 in hilly & mountainous terrain

W is width of pavement

(3) empirical formula as per IRC

- plain & rolling terrain formula
- mountainous or steep terrain

(3) Empirical formula as per IRC

(i) plain & rolling terrain

$$L_s = \frac{2.7V^2}{R} \quad \text{or} \quad \frac{2.7\,(\text{Velocity in km})^2}{\text{Radius of Xs curve}}$$

(ii) Mountainous or steep terrain

$$L_s = \frac{v^2}{R} \quad \text{or} \quad \frac{\text{velocity}^2}{\text{radius}}$$

- Teeno me se criteria me se kaunsa large finally

among all 3 criteria (maximum) length is considered

① ② ③ } ↑ max

Shift of curve

- formula
- what is L_s

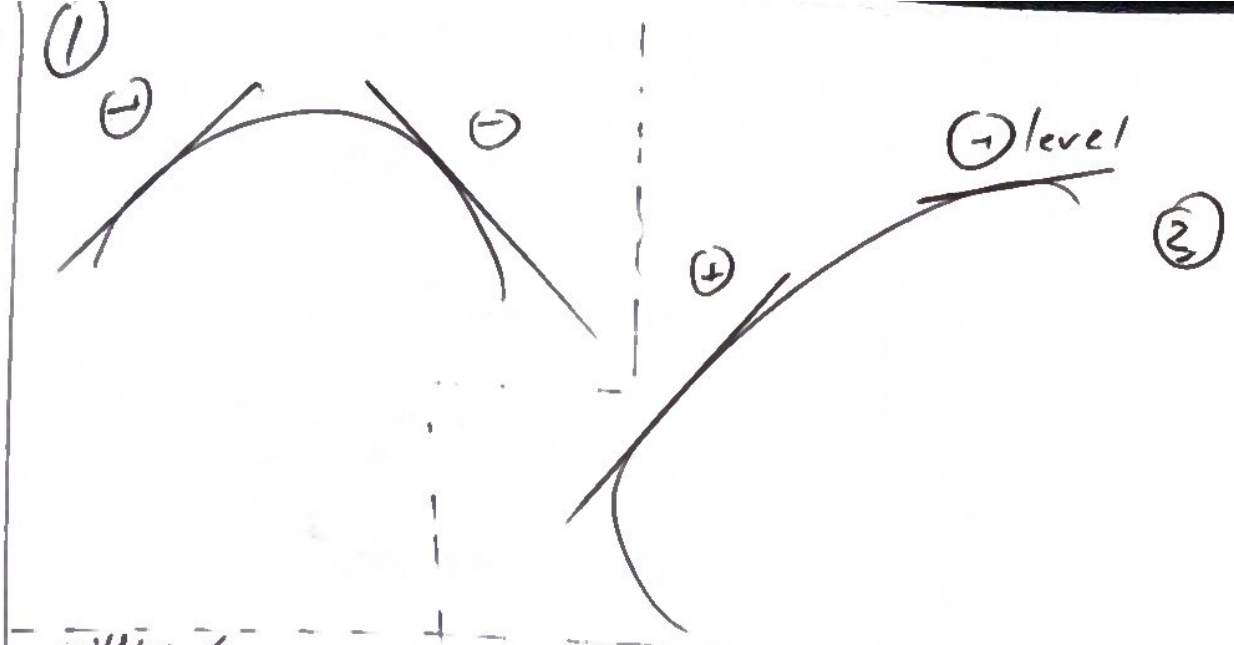
Shift of curve

$$S = \frac{L_s^2}{24R} \quad \text{or} \quad \frac{(\text{Length of transn curve})^2}{24 \times \text{Radius}}$$

4.5 • Vertical curve

S: ② Summit curve

- Diagram



- kab kab detch

- + -
- + mild +
- + 0
- - steeper -

- length of summit curve (Design)

Summit curve is provided when

① +ve grade [ascending grade] meets -ve grade [descending grade]

② +ve grade meets another mild +ve grade

③ When +ve grade meets level sketch

④ When -ve grade meets steeper -ve grade

length of summit curve design.

1. Design criteria based on SSD.

- length of summit curve design
  - Based on ssd
    - assume $L \gg$
      - formula
        - H kyak
        - h kyak
        - h_1 kyak
      - formula when std values are taken
    - How to calculate geometric difference in grades or deviation angle.

(a) assume $L \gg SSD$

$$L = \frac{NS^2}{\left(\sqrt{2H} + \sqrt{2h}\right)^2}$$

$$= \frac{\text{deviation angle} \times SSD^2}{\left[\sqrt{2 \times \text{drivers eye ht}} + \sqrt{\text{obstruct}^n \text{ ht}}\right]^2}$$

$$L = \frac{NS^2}{\left[\sqrt{2 \times 1.2} + \sqrt{2 \times 0.15}\right]^2}$$

$$L = \frac{NS^2}{4.4}$$

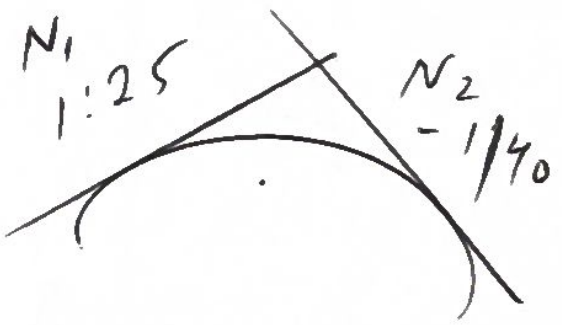
N is deviation angle or algebraic difference in grades



$$N = |+3 - (-5)|$$

$$= 8\%$$

$$= 8/100$$



$$N = |N_1 - N_2|$$

$$= \left|\frac{1}{25} - \left(\frac{-1}{40}\right)\right|$$

$$= \left|\frac{1}{25} + \frac{1}{40}\right|$$

$$= 0.065$$

length of summit curve

- OSD
  - assume L > OSD
    - formula
    - formula for std value
    - Check
      - Pass
      - fail
        - formula
        - formula using std values

① Design criteria based on OSD (also valid for ISD)

(a) assume $L > OSD$ or ISD

$$L = \frac{NS^2}{\left[\sqrt{2H} + \sqrt{2H}\right]^2}$$

$$L = \frac{NS^2}{9.6}$$ Yahan S ko OSD ya ISD lenge.

Check

— Pass OK ②

→ Provide the calculated L

— fail Not ok (b) $L \not> OSD$ or ISD
$L < ISD/OSD$

$$L = 2S - \frac{\left[\sqrt{2H} + \sqrt{2H}\right]^2}{H}$$

$$= 2S - \frac{9.6}{N}$$

Minradius of parabolic summit curve formula.

Note minimum radius of parabolic summit curve

$$R = \frac{L}{N}$$

angle = $\frac{arc}{Radius}$



U-9 Summit curve

- Best shape?
- eqn of sq pbla

Best shape for summit curve is square parabola.

Equation of square parabola $y = \frac{Nx^2}{2L}$

Valley curve

- alternate names
- Best shape for valley curve all names.
- use
- Design criteria
  - HLD
  - M comfl
  - Cross drng ctrl
  - Gen aes

Vertical curve (Valley curve)

. Sag curve / Concave crve

. Best shape for valley curve is cubic parabola / easement curve / froudes transition curve

. use: purane wale me jahan +ve tha woh -ve aur -ve tha woh +ve ho jayega / summit ke opposite

- Design criteria

① . Vehicle headlight distance

② . Motorist comfort.

③ Cross drainage control @ lowest pt of valley

④ General aesthetics

Design of valley curve

① Comfort criteria

- kyd discomfort hota h?

- formula

- C value ?

- Std formula wit C included

② HSD

- assume L > HSD

L formula

- Bkyq h?

- Std formula

Design:

① Comfort criteria

. In case of valley curve the pressure on engine or spring are more due to centrifugal force and gravity or weight of vehicle all acting downward. Hence it causes discomfort to driver.

$$L = 2 \times \sqrt{\frac{N v^3}{C}}$$

put $C = 0.61 \text{ m/s}^3$ $\quad v = 0.278\,V$ ← we get

$$L = 0.38 \sqrt{N V^3} \quad \text{kmph}$$

② Design based on headlight sight distance (also valid for SSD)

ⓐ assume $L > HSD$

formula $L = \dfrac{N S^2}{[2h_1 + 2 \tan\beta]}$

h_1 = head light beam L = .75

β = tan 1° = 0.017

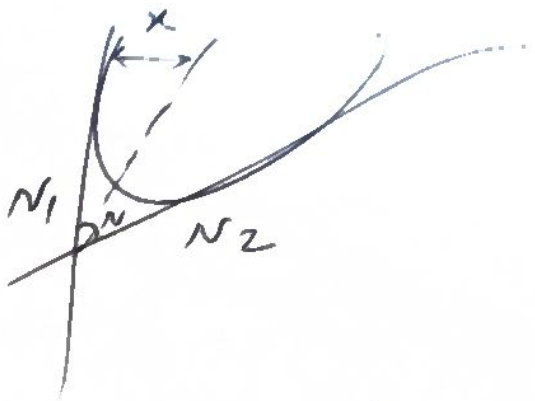
$$L = \frac{N S^2}{2 \times 0.75 + 2 \times 0.017} = \frac{N S^2}{1.5 + 0.034 S}$$

∝ std formula

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Centrifugal ratio

N_1 is gradient of initial tangent



N is deviation angle; x is location of deepest or deepmost point from initial tangent.

L is length of valley curve

location of deepmostpt = $\frac{\text{gradient of initial tangent} \times \text{length of valley curve}}{\text{Deviation angle}}$

- alternate name — Impact factor or centrifugal ratio
- kya h z.
- formula & alternate formula — . It is the ratio of max centrifugal force to the weight of vehicle.
  . $I = P/W$ or $I = 1.6NV^2/L$
- Impact factor kisse zyada nahi hoskta — . Impact factor cannot be more than 17% as per IRC

Vertical curve design

- ① HSD
  - assume $L > HSD$
    - Check pass
    - Check fail
      - formula
      - std formula

Check

- if pass OK $L > HSD$ ② L ✓
- if fail Not OK ⑤

⑤ $L < HSD$ or $L \neq HSD$

$$L = 2S - \frac{(2h_1 + 2S\sin\beta)}{N}$$

$$L = 2S - \frac{1.5 + 0.034S}{N}$$

- Why is OSD not a problem in Vertical curve.

Note OSD: overtaking sight distance is not a problem for valley curve because in daytime visibility is clear & during nighttime headlights ~~distance~~ can be seen from a considerable distance.

- Position of deepmostpt in valley curve formula.

. The location of deepest or deep most point in valley curve $x = \frac{N_1 L}{N}$

5

Curve resistance
- kya h?
- kin vehicles me hota h
- Hills pe kaise hota h
- formula
  - T kyah
  - θ kyah
- Tractive force definition

Setback, Curve resistance & Grade compensation

Class 5

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Curve resistance

Curve resistance is the less in tractive force / traction effort due to turning of vehicle on a horizontal curve.

- Curve resistance develops only in vehicles having rear driving wheels
- Curve resistance is acute in hillroads

$\Delta T = T - T\cos\theta$

T is actual tractive force

θ is turning angle

- Tractive force: force needed to overcum the resistance caused by friction when two bodies slide on eachother

Setback

- alternate name
- Defn
- approximate formula
- factors affecting
  - 1
  - 2
  - 3
- Rational formula

Setback / Setback distance M

• Clearance

. It is the distance required from the centreline of horizontal curve to an obstruction on the inner side of curve to provide adequate sight distance @ a horizontal curve.

$M = L^2/8R$ — approximate formula for M

or $\frac{\text{Length of curve}}{8 \times \text{Radius}}$

Setback distance M $\frac{S^2}{8R}$

or $\frac{\text{sight distance desired}^2}{8 \times \text{Radius of curve}}$

factors affecting setback distance

1. Sight distance
2. length of curve
3. Radius of horizontal curve.

Rational formula

6.3 Jelback

Rational formula
- ① L ≥ S
  - Single lane
  - multi lane
    - d
      - key
      - plan
    - diagram
- L < S
  - formula for α

① L ≥ S

↙ single lane

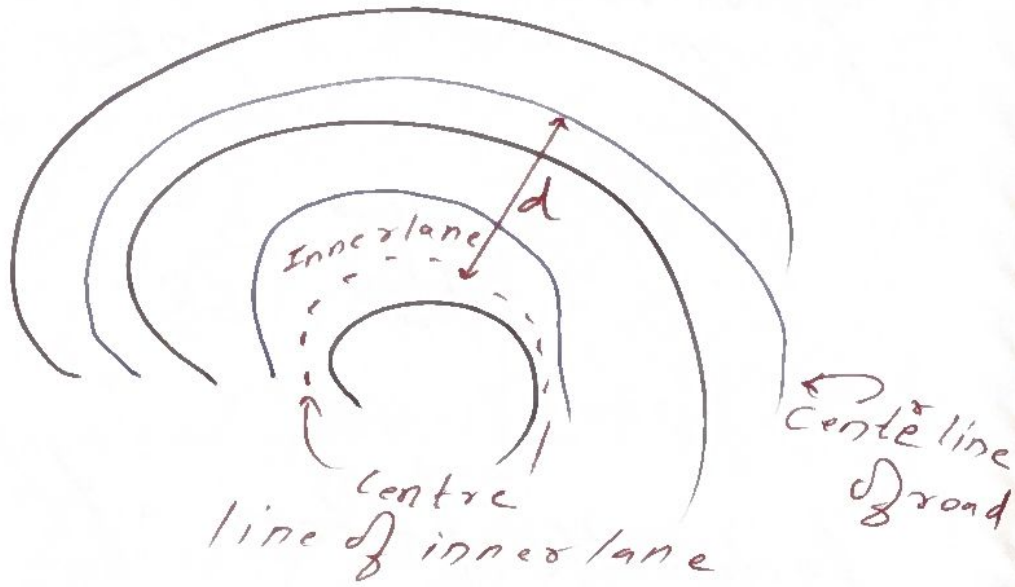
$$M = m = \left(R - R\cos\frac{\alpha}{2}\right)$$

R

if multilane

$$M = m = R - (R-d)\cos\frac{\alpha}{2}$$

d is distance between centre line of road & centre line of inner lane.

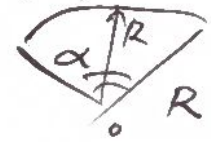


(b) When L < S (no time)

Single lane $M = \left(R - R\cos\frac{\alpha}{2}\right) + \left(\frac{S-L}{2}\right)\sin\left(\frac{\alpha}{2}\right)$

multilane $M = R - (R-d)\cos\frac{\alpha}{2} + \left[\frac{S-L}{2}\right]\sin\left(\frac{\alpha}{2}\right)$

$\frac{\alpha}{2} = \frac{180\ S}{2\pi R}$ ← single lane



Setback
- Rational formula for multilane
  - L&L (not imp)

$$\frac{\alpha}{2} = \frac{180 S}{2\pi (R-d)} \leftarrow \text{for multilane}$$

S is sight distance
R is Radius of curve fm centre line of road.
m is clearance or setback.

Grade compensation

- grade compensn
  - Defn, kab dete h?
  - kab naidenge
  - formula.

The reduction given to the gradient when a gradient comes along with horizontal curve as per IRC grade compensation should be provided

Grade compensation should not be given to the gradient less than or flatter than 4%. G = 4%.

formula G = gradient or grade or longitudinal slope (%)

GC = Grade compensation.

$$GC_{(\%)} = \left[\frac{30 + R}{R}\right]\%$$

$$\left[\frac{75}{R}\right]$$

R is rad of curve in m